Security Valuation

Instruments

- Debt Instruments
  - Coupon instruments
  - Treasury instruments
  - Zero coupon instruments
  - Flexi coupon instruments
  - Deep discount instruments
- Equity Instruments
  - Common equity instruments
  - Preferred equity instruments
  - Convertible instruments
  - Derivative instruments

Coupon Instruments

Symbol	Series	Coupon Rate	Face Value	LTP	% Change	Volume (Shares)	Value (₹ Crores)	CREDIT RATING AGENCY	CREDIT RATING	Maturity Date
850HMDA40	N0	8.5	1,00,000.00	1,02,910.96	2.91	100	1.03	FICH, ACUITE	AA/Stable, AA+/Stable	15-Jun-40
830NHAI27	N2	8.3	1,000.00	1,079.00	0.1	9,021	0.97	-	-	25-Jan-27
96IIFL28A	NF	9.6	1,000.00	1,001.00	-0.17	3,703	0.37	CRISIL	AA/Negative	24-Jun-28
915APSBC35	N0	9.15	1,00,000.00	1,05,717.94	0.17	34	0.36	IRRPL, ACUITE	AA/Stable, AA/Stable	30-Nov-35
915APSBC29	N0	9.15	1,00,000.00	1,03,030.97	0.91	30	0.31	IRRPL, ACUITE	AA/Stable, AA/Stable	30-Nov-29
115AFIL28A	N0	11.5	10,000.00	9,876.90	-0.41	54	0.05	ACUITE	A-/Stable	14-Jul-28
1025SCL34	Z8	10.25	1,000.00	1,040.00	0	501	0.05	CRISIL	AA/Stable	26-Mar-34

Source: https://www.nseindia.com/market-data/bonds-traded-in-capital-market.

Valuation of Coupon Instruments

- Running yield

$$YIELD_r = \frac{C}{CP}$$

- Simple yield to maturity

$$YIELD_s = \frac{C}{CP} + \frac{P - CP}{n \times CP}$$

- Yield to maturity

$$YTM = \sum_{i=1}^{n} \frac{C_i}{(1+k)^i} + \frac{P}{(1+k)^n}$$

➢ C is the coupon amount,
➢ P is the face value of the bond,
➢ CP is the clean price,
➢ n is the number of years to maturity
➢ K is the discount rate

IIFL Finance Limited (INE530B01024)

- Market price: Upper band: 1,201.20; Lower band: 800.80 52 week High: 1,010.00; 52 week Low: 890.00
- Coupon rate: 9.6%
- Current market Price: Closing price: 1001
- Running yield: 9.59%
- Simple Yield to Maturity: 9.54%
- Yield to Maturity: 9.54%
- Theoretical Value: 1065.04
- Accrued Interest: 96x(37/365)
- Dirty Price: 1065.14

$$YIELD_r = \frac{96}{1001}$$

$$YIELD_S = \frac{96}{1001} + \frac{1000 - 1001}{2 \times 1001}$$

$$Accrued\ Interest = \frac{C}{m} \times \left(\frac{d_{xi} - d_{xc}}{T_d}\right)$$

Note: dxi: Number of days since the last coupon payment; dxc: Number of days between book closure dates; Two year zero coupon yield curve rate: 6.05% (k)

Equity Valuation: Multiplier Method

- Earnings Valuation
- Revenues Valuation
- Cash Flow Valuation
- Economic Value Added
- Accounting Value
- Yield Valuations
- Member’s Valuation

Multiplier Methods

- **Price-Earnings Ratio**
  - P/E multiplier
  - P/E Growth rate = P/E Multiplier / Growth Rate
  - P/E relative = (share P/E) / (Index P/E)
- **Price to Sales Ratio**
  - Market capitalization / One year total revenue
  - Market Capitalization = (Shares Outstanding * Current Share Price) + Current Long-term Debt
- **Price to Book value Ratio**
  - Market price / Book value per share

Intrinsic Value of Share

Current Market Price: INR 3787

		INR Crore	INR Crore	INR Crore	INR Crore
Company Name	Date	Sales	Equity dividend	Borrowings	Total assets
	2024	2,21,113	34.20	1,16,322	3,39,602
	2025	2,55,734	33.90	1,32,409	3,79,114
L&T	2026	2,85,874	37.41	1,25,497	4,52,383

		INR Crore	INR	Number	INR
Company Name	Date	PAT	NSE Average price	Outstanding shares	EPS on accounting year end
	2024	15,547	3590.35	1375768535	95
	2025	17,673	3625.72	1375768535	109.35
L&T	2026	18,954	4081.03	1375768535	116.92

Intrinsic Value (P/E Multiplier)

	Prior to Previous Year	Previous year	Latest Year	Forecast Year (Growth rate)	Forecast Value
NSE Average price	**3590.35**	**3625.72**	**4081.03**		
EPS on accounting year end	**95.00**	**109.35**	**116.92**		152.1187131
PAT (Crore)	**15547.00**	**17673.00**	**18954.00**	0.1041476	20928.0139
Outstanding shares (Number)	**1375768535**	**1375768535**	**1375768535**		
Index P/E	**26.4**	**29.5**	**30.06**	0.0670691	

	Prior to Previous Year	Previous year	Latest Year	Forecast Year (Average Multiple)	Forecast Price
Index P/E	26.4	29.5	30.06	**29.26**	**4451.50**
Earning Valuation					
P/E	37.79	33.16	34.90	**34.80**	**5294.25**
P/E Growth (%)	5.63	4.94	5.20	**5.19**	**4292.80**
P/E Relative	1.43	1.12	1.16	**34.94**	**5314.33**

Intrinsic Value (PSR Multiplier)

	Prior to Previous Year	Previous year	Latest Year	Forecast Year (Growth rate)	Forecast Value
NSE Average price	3590.35	3625.72	4081.03		
Outstanding shares (Number)	1375768535	1375768535	1375768535		
Borrowings (Crore)	116322	132409	125497	0.0386895	130352.4163
Sales (Crore)	221113	255734	285874	0.1370516	325053.5019

	Prior to Previous Year	Previous year	Latest Year	Forecast Year (Average Multiple)	Forecast Price
Revenue Valuation					
Market Cap + debt	6102710559637	6312241492720	6869522664391		807515.97
Revenue	2211130000000	2557340000000	2858740000000		
PSR	2.760	2.468	2.403	2.48	4922.07

Intrinsic Value (P/B Multiplier)

	Prior to Previous Year	Previous year	Latest Year	Forecast Year (Growth rate)	Forecast Value
NSE Average price	3590.35	3625.72	4081.03		
Total assets (Crore)	339602.000	379114.000	452383.000	0.1541653	522124.77
Outstanding shares (Number)	1375768535.000	1375768535.000	1375768535.000		

	Prior to Previous Year	Previous year	Latest Year	Forecast Year (Average Multiple)	Forecast Price
Accounting Valuation					
Equity Worth	3590	3626	4081		
Book Value	2468.45	2755.65	3288.22		
Price to Book Value	1.45	1.32	1.24	1.30	4939.57

Adjustments to Valuation

- Multiples from similar firms / competitive firms
- Adjust multiples for
  - Liquidity (10% to 30%): Firm is closely held with very little active trading, then a non-marketability discount would be subtracted from the value.
  - Size of the shareholding also needs to be adjusted.
  - Control premium indicates stock is not freely traded. Controlling interest assumes control over minority interest. Minority interest discount represents reduction in value for the absence of control.
  - Public Vs. Private firms (20% to 30%): Enterprise Value /EBITDA in public and private transactions adjusted such that transactions are comparable.

DCF Methods

- **Economic Value Addition**
  - [(Net operating profit after tax / Capital employed) – Weighted cost of capital] x Capital employed
- **Market Value Addition**

$$V = \sum_{t=1}^{n} \frac{EVA_t}{(1+d)^t}$$

- **Cash flow valuation**

$$V = \sum_{t=1}^{n} \frac{CF_t}{(1+d)^t}$$

- **Dividend Valuation**
  - Expected dividend / cost of capital
  - Expected dividend / (cost of capital - growth rate)
- **Member's Valuation**
  - Subscription value multiplier

Cash Flow Valuation: Growth Rates

- Normal growth rates
  - One Year Growth rate: ROE x retention rate
- Sustainable growth rates
  - Growth in EPS: ROE x (1 – Dividend Payout Ratio)
  - Growth in Investment: ROIC x (1-Reinvestment Rate)
    - $ROCI = \frac{EBIT \times (1 - tax\ rate)}{Investment}$: Net operating profit less adjusted taxes / investment
  - Long-term Growth: (1+growth in units) x (1+inflation) – 1
  - Growth in free cash flows = (Capital Expenditures/Depreciation – 1) x Depreciation Rate
- Free Cash Flow = ROIC x Investment – Capital Expenditure + Depreciation
- Free Cash Flow = Net operating profit less adjusted tax – Capital Expenditure + Depreciation

Cash flow Valuation

- Gordon's Model: $V = \frac{D_1}{(k-g)}$; $V = \frac{CF_1}{(k-g)}$
- Discount rate k: Cost of equity
  - P = D1/(k-g); g = ROE * (1-Dividend Payout ratio (DPO)) or DPO = 1 - g/ROE
  - P/E = Dividend(D) / Earning(E)/(k-g); Substituting for D/E = DPO; P/E = (1-g/r)/(k-g)
  - (k-g) = (1-g/r)/(P/E); k = (1-g/r)/(P/E) + g
  - P = E/(k-g) x (1-g/r)
- Terminal Value: $\frac{CF_1 \times (1+g)}{(WACC-g)}$
- Firm Value = Explicit forecasts + Terminal value + Listed investments + Other investments + Cash holdings – Bank and other debt – Minority holdings

Intrinsic Cash Flow Valuation

- Growth Rate

	Rate
ROE	13%
Dividend payout ratio	32%
g	8.5%

- Cost of Debt

Debt	
Debt Interest	10,056
Borrowing (Balance Sheet)	1,25,497
Debt rate	8.03%

- Cost of Equity

beta	0.79
Market return	13%
Risk free rate	6.72%
Equity Value	1,09,290
equity rate	11.33%

- Cost of Capital

WACC	0.0955866

- Firm Value

Dividend	37.4144
Value (Dx(1+g)/(k-g)	**3834.52**